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Reversal asymmetry in Mesoproterozoic overprinting of the 1.88-Ga Gunflint Formation, Ontario, Canada: non-dipole effects or apparent polar wander?

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Abstract

Eighty-two palaeomagnetic samples of calcareous and jaspilitic grainstones (iron-formation or 'taconite') and chert carbonate were collected from the 1.88-Ga Gunflint Formation at 22 sites in the Thunder Bay area, Ontario. Twenty clasts of Gunflint taconite also were sampled from the basal conglomerate of the overlying Mesoproterozoic Sibley Group. Anisotropy of magnetic susceptibility measurements indicate the Gunflint Formation in the sampling area has not experienced regional dynamic metamorphism. Analyses by variable-field translation balance and X-ray diffraction show that the predominant magnetic mineral is hematite but a small amount of magnetite also is present in some samples. Altogether, 213 Gunflint specimens and 59 Sibley conglomerate specimens were subjected to stepwise thermal demagnetisation and 74 Gunflint specimens to stepwise alternating-frequency demagnetisation. The following components were isolated for the taconites:

- Gunflint magnetite: normal declination $D=293.4°$, inclination $I=30.8°$, $\alpha_{95}=7.2°$, $n=21$; reverse $D=86.7°$, $I=-54.6°$, $\alpha_{95}=5.8°$, $n=29$.
- Gunflint hematite: normal $D=243.6°$, $I=23.6°$, $\alpha_{95}=6.0°$, $n=11$; reverse $D=70.3°$, $I=-51.4°$, $\alpha_{95}=3.2°$, $n=79$.
- Sibley clasts magnetite: normal $D=282.7°$, $I=33.4°$, $\alpha_{95}=7.6°$, $n=20$.
- Sibley clasts hematite: normal $D=254.5°$, $I=56.2°$, $\alpha_{95}=8.4°$, $n=13$; reverse $D=110.6°$, $I=-55.7°$, $\alpha_{95}=8.3°$, $n=11$.

None of these sets passed the reversal test, with the normal component generally being the shallower. Fold tests were negative or inconclusive and the conglomerate test also was negative. Chert carbonate at one other site appears to have acquired a remanence carried by magnetite ($D=97.3°$, $I=-78.2°$, $\alpha_{95}=6.3°$, $n=18$) prior to folding related to Keweenawan (1.1 Ga) Logan diabase sill emplacement. Most of the components we identified match components for Keweenawan sills, volcanic rocks, intrusions and baked contact rocks in the Thunder Bay area, indicating that Keweenawan magmatism caused widespread chemical remagnetisation of the Proterozoic country rock in our sampling area. Although others have argued that asymmetry was a feature of the Keweenawan geomagnetic field, the declinations of our Gunflint and Sibley hematite and magnetite components are different, suggesting that the components were acquired at

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significantly different times. We conclude that the reversal asymmetry shown by our Gunflint and Sibley data may best be ascribed to apparent polar wander during Keweenawan times.
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1. Introduction

Elucidating the palaeomagnetism of Palaeoproterozoic iron formations is notoriously difficult. The primary mineralogy of the iron formations can be complex, and spurious components of magnetisation may be produced through thermo-chemical alteration of minerals with heat treatment. Moreover, burial diagenesis and metamorphism commonly have produced secondary magnetite carrying overprint components that further complicate the palaeomagnetic signature (e.g. Schmidt and Clark, 1994). Partly because of such difficulties, most palaeomagnetic studies of iron formations have focussed on the identification of components related to the development of secondary iron ores (e.g. Symons, 1967; Porath, 1967; Li et al., 1993). Other, early studies seeking depositional palaeolatitudes, as for the Gunflint Formation in Ontario (Symons, 1966) and the Sokoman Formation in Québec (Seguin, 1976), did not employ modern palaeomagnetic analytical techniques or conduct complete or stepwise demagnetisation and field tests. Consequently, few if any reliable data are available on the original components of magnetisation of Palaeoproterozoic iron formations and, as cautioned by Trendall (1997), palaeomagnetic data for such iron formations should not be uncritically interpreted in terms of palaeolatitudes of deposition. The finding of a palaeolatitude of $13.0 \pm 1.5^\circ$ for iron formation from the Frere Formation (1.9–1.8 Ga) in Western Australia (Williams et al., in press) is the best-constrained palaeolatitude directly determined for a Palaeoproterozoic iron formation.

The present study of the Gunflint Formation (1.88 Ga) in the Thunder Bay area of northwestern Ontario extends our investigation of the palaeomagnetism of Palaeoproterozoic iron-formations that aims to illuminate their palaeogeography and metamorphic history.

2. Geological setting

The Gunflint Formation of the late Palaeoproterozoic Animikie Group (Moorhouse, 1960; Morey, 1983) forms a belt of chemical, clastic and volcaniclastic sedimentary rocks 100–160 m in thickness (average = 120 m) that extends for 175 km along the Gunflint Range from northeastern Minnesota to the Thunder Bay area, Ontario (Fig. 1). The formation was deposited on peneplaned Archaean crystalline rocks at the southwestern margin of the Superior Province, and is famous for its Superior-type granular peloidal and oolitic iron formation or 'taconite' (Goodwin, 1956; Floran and Papike, 1975) and well-preserved microfossils and stromatolites (Awramik, 1976).

Fralick and Barrett (1995) divided the Gunflint Formation into two members (Fig. 2) based on the lithostratigraphic subdivisions of Goodwin (1956). The lower member typically has a basal conglomerate with algal and oncolitic chert, succeeded by cherty calcareous grainstones that to the east and west pass into jaspilitic grainstone facies (the Lower West Taconite and Lower East Taconite). The upper member has a stratigraphy similar to that of the lower member: algal chert is followed by a succession of cherty calcareous grainstones and banded chert carbonate that passes westward into jaspilitic grainstone facies (Upper Taconite). Extensive tuffaceous shales with local accretionary lapilli are interbedded with the grainstone succession. A limestone dolomite unit occurs at the top of the upper member and is conformably overlain by shale of the Rove Formation. Fralick and Barrett (1995) interpreted the chemical sediments of the Gunflint Formation as a wave- and tide-dominated inner-shelf succession.

Faure and Kovach (1969) obtained a Rb–Sr whole-rock age of 1635 ± 24 Ma ($\lambda^{87}\mathrm{Rb} = 1.39 \times 10^{-11}$ year^{-1}) for six samples of the Gunflint Formation (chert, lithic tuff, argillite, iron carbonate, cherty iron

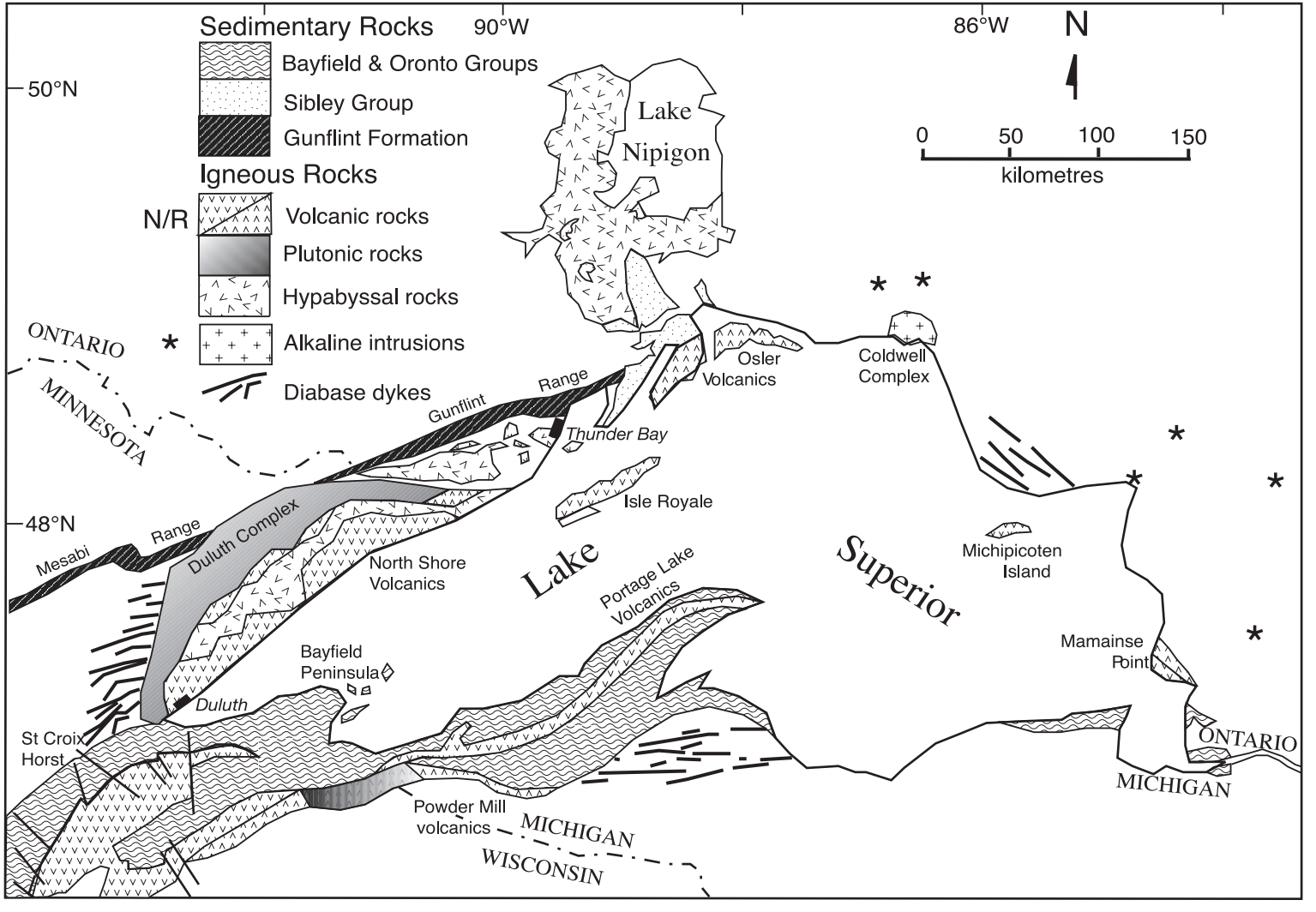


Fig. 1. Generalised geological map of the Lake Superior region, showing the regional distribution of the Gunflint Formation, the Sibley Group, and Keweenawan igneous and sedimentary rocks. N indicates normal polarity and R indicates reverse polarity. Adapted from Ojakangas et al. (2001) and Cheadle (1986).

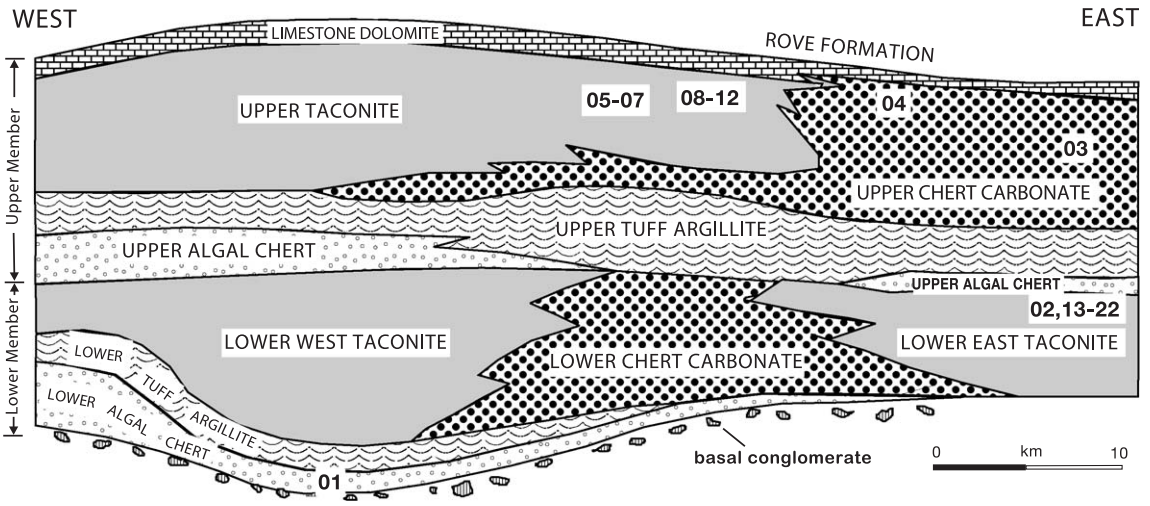


Fig. 2. Schematic section through the Gunflint Formation along the Gunflint Range, Ontario, showing the two members of Fralick and Barrett (1995) and the lithostratigraphic subdivisions of Goodwin (1956). The Gunflint Formation is 100–160 m thick (average = 120 m). The numbers mark Gunflint palaeomagnetic sites (GU01–GU22) for the present study.

carbonate, cherty carbonate) and one sample of Rove Formation argillite, a figure that converts to 1600 ± 24 Ma (λ^{87}Rb = 1.42 × 10^{-11} $year^{-1}$). Franklin (1978b) gave a Rb–Sr whole-rock age of 1556 ± 64 Ma (λ^{87}Rb = 1.47 × 10^{-11} $year^{-1}$; Wanless and Loveridge, 1978) for argillite from the Rove Formation, which converts to 1610 ± 64 Ma (λ^{87}Rb = 1.42 × 10^{-11} $year^{-1}$). Fralick et al. (2002) obtained a U–Pb age of 1878.3 ± 1.3 Ma for a population of euhedral volcanogenic zircons from reworked volcanic ash in the upper member of the Gunflint Formation near Thunder Bay. This age, which Fralick et al. (2002) regarded as nearly synchronous with Gunflint deposition, is analytically indistinguishable from a U–Pb zircon age of 1874 ± 9 Ma obtained for rhyolite coeval with the Negaunee Iron formation in northern Michigan (Schneider et al., 2002). Hence, the Rb–Sr whole-rock ages for the Gunflint and Rove formations, which are regarded as conformable (Franklin and Kustra, 1972; Fralick et al., 2002), appear to be at least 200 million years younger than the depositional age.

The Animikie Group was folded during the Penokean Orogeny (Riller et al., 1999), which caused deformation and metamorphism of Palaeoproterozoic and Archaean rocks along the southern margin of the Superior Province 1.89–1.83 Ga. According to Schneider et al. (2002), iron formation accumulated within an extensive foredeep in the western Lake Superior region during arc accretion from the south. In the Thunder Bay area, the Animikie Group has a gentle (< 12°) southeastward regional dip and strata are relatively undeformed and of subgreenschist facies.

The Animikie Group is unconformably overlain by the Sibley Group, a mixed clastic–carbonate red bed succession with a total thickness of 420 m and averaging 150 m (Franklin et al., 1980; Ojakangas and Morey, 1982; Cheadle, 1986). At Pass Lake 40 km northeast of Thunder Bay, a polymict conglomerate of the Pass Lake Formation at the base of the Sibley Group lies unconformably on the Rove Formation and contains abundant subrounded pebbles and cobbles of jaspilite and taconite derived from the Gunflint Formation. The Sibley Basin is envisaged as a north–south-trending continental rift (Franklin et al., 1980) or sag (Cheadle, 1986).

The age of the Sibley Group is problematic. Franklin (1978a) gave a Rb–Sr whole-rock age of 1294 ± 33 Ma (λ^{87}Rb = 1.47 × 10^{-11} $year^{-1}$; Wanless and Loveridge, 1978) for dolomitic mudstone and shale. Franklin et al. (1980) converted this age to 1339 ± 33 Ma (λ^{87}Rb = 1.42 × 10^{-11} $year^{-1}$), which is taken as the age of the Sibley Group (Cheadle, 1986). However, the Rb–Sr whole-rock shale ages of 1600 ± 24 and 1610 ± 64 Ma for the Gunflint and Rove formations (see above) appear to be at least 200 million years too

young, raising the possibility that the whole-rock age of 1339 ± 33 Ma for the Sibley Group also is a minimum. Davis and Sutcliffe (1985) obtained a U–Pb zircon age of 1536.7 + 10/ − 2.3 Ma for a quartz-feldspar porphyry at English Bay, Lake Nipigon, that was seen to grade into porphyritic fragmental rocks intercalated with quartz arenite possibly coeval with lowermost Sibley strata. They concluded that the age for the porphyry may give the age of deposition of the basal Sibley Group in the Lake Nipigon area. The age of the Sibley Group is further discussed in Section 7.4.

The comparative stratigraphy and magnetic polarity of the late Mesoproterozoic Keweenawan Supergroup in the Lake Superior region is shown in Fig. 3. Keweenawan magmatism was related to the Midcontinent Rift System, and U–Pb geochronology indicates two major magmatic phases at ~ 1108–1105 and 1100–1094 Ma (Davis and Green, 1997). In the western Lake Superior region, the oldest ages determined for Keweenawan igneous rocks are 1108.2 ± 0.9 Ma for the basaltic and peridotitic Logan sills (Davis and Sutcliffe, 1985; Davis and Green, 1997) that cut the Archaean, Palaeoproterozoic and Mesoproterozoic rocks. Numerous Logan sills and sheets cross-cut Animikie Group strata at low angles in the Thunder Bay area and range in thickness from 5 to >60 m (Moorhouse, 1960); the sills crop out boldly, capping cuestas, buttes and mesas. Plateau volcanics of the ~ 2.75-km-thick Osler Group (Green, 1977; Ojakangas et al., 2001) were extruded in the Thunder Bay–Lake Nipigon area penecontemporaneously with emplacement of the Logan sills. A rhyolite from the

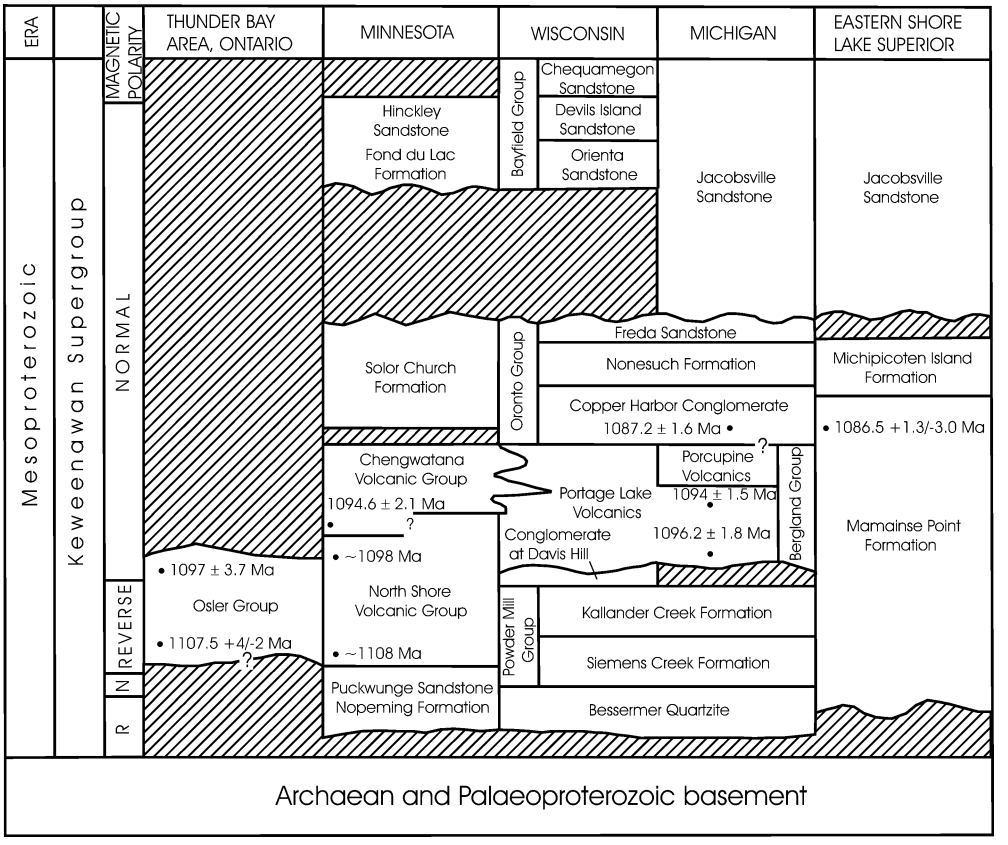


Fig. 3. Stratigraphy and magnetic polarity of the late Mesoproterozoic Keweenawan Supergroup in the Lake Superior region (modified after Ojakangas et al., 2001). U–Pb and U–Th–Pb ages are from Davis and Sutcliffe (1985), Palmer and Davis (1987), Davis and Paces (1990), Paces and Miller (1993), Davis and Green (1997) and Zartman et al. (1997).

Osler Group gave a U–Pb age of 1097.6 ± 3.7 Ma and a porphyry, probably a flow, from the base of the Osler Group a U–Pb age of 1107.5 + 4/ − 2 Ma (Davis and Sutcliffe, 1985). Other Keweenawan igneous rocks in the Thunder Bay area include northeast-trending gabbro dykes that cut the Logan sills, and several gabbro stocks, some of which intrude the Osler Group volcanics (Franklin and Kustra, 1972; Smith and Sutcliffe, 1987). The gabbro stocks crop out 20–60 km from our sampling sites.

Intrusion of the Logan sills and sheets caused local disruption of the Proterozoic country rock. Moorhouse (1960) and Purucker (1982, 1983) noted the close spatial relationship between a Logan sill and local folding and brecciation of Gunflint banded chert carbonate south of Blende Lake, and suggested that deformation resulted from sill emplacement. Local folding, faulting and shearing of the Sibley Group adjacent to the Logan sills occurred during two episodes of shear strain related to sill intrusion (Antonellini and Cambray, 1992).

Burke and Dewey (1973) proposed that the Lake Superior region marks the site of an ancient triple junction, with two of the arms being the southwest- and southeast-trending parts of the Midcontinent Rift System. Pursuing this idea, Franklin et al. (1980) suggested that the Sibley Group is situated, and possibly was deposited, in the 'failed arm' that radiates northward from the Lake Superior region. Other workers (Davidson, 1982; Ojakangas and Morey, 1982; Cheadle, 1986) were unconvinced that the Sibley Group occupies a failed arm of the Midcontinent Rift System because of the age discrepancy, the apparent lack of igneous activity during Sibley deposition, and doubts concerning the rift setting of the Sibley Basin.

3. Previous palaeomagnetic studies

3.1. Gunflint Formation

Symons (1966) studied the palaeomagnetism of the chert carbonate facies of the Gunflint Formation in the Thunder Bay area. Some of the sampling sites were in close proximity to Logan diabase sills. He gave results for 20 samples, 2 of which were subjected to stepwise alternating-field (AF) demagnetisation to 70 mT and the remainder to bulk AF demagnetisation at 70 mT. Symons (1966) found a mean direction of $D = 192°$, $I = 79°$ ($\alpha_{63} = 5°$), yielding a palaeomagnetic pole position of 28°N,266°E. He concluded that the remanence was a primary chemical remanent magnetisation (CRM) acquired at the time of deposition and that the Gunflint Formation was deposited at a polar palaeolatitude. However, no field test was executed to constrain the timing of magnetisation.

Halls (1979) obtained data for two samples of Animikean argillite that overlies cherty iron formation, collected from directly below Keweenawan basalt on the Slate Islands 160 km east of Thunder Bay. He suggested a possible primary origin for the Animikean direction obtained (in situ $D = 49.2°$, $I = -24.0°$; bedding dip-corrected $D = 61.0°$, $I = -9.6°$, $\alpha_{95} = 33.3°$), although he observed that 'the samples come from the uppermost 3 m of the section beneath the lowermost Keweenawan flow, where the normally grey argillite becomes red.' Halls (1979) noted that the bedding dip-corrected direction yields a pole position close to that found by Robertson (1973a) for the Sibley Group.

Atuanya (1982) carried out mostly bulk AF demagnetisation at 20 mT and bulk thermal demagnetisation at 550 °C of Gunflint samples from 47 sites in the Thunder Bay area between Whitefish Lake in the southwest and Loon Lake in the northeast. Three components were isolated by point density contour plots. Component A ($D = 327.0°$, $I = 76.8°$, $\alpha_{95} = 3.0°$; pole position of 63.6°N,236.2°E) was interpreted as primary; a fold test was inconclusive, although a positive conglomerate test was claimed following AF demagnetisation to 20 mT of an unspecified number of jasper clasts at one site. Component B ($D = 212.3°$, $I = 68.4°$, $\alpha_{95} = 8.4°$; pole position of 13.4°N,251.1°E) and component C ($D = 135.8°$, $I = 42.3°$, $\alpha_{95} = 10.1°$; pole position of 7.5°N,310.8°E) were interpreted as overprints related to the Hudsonian (Penokean) Orogeny. Subsequently, Atuanya et al. (1983) placed the B pole at 7.5°N,320°E, which they regarded as indicating a Hudsonian metamorphic overprint.

Purucker (1983) used stepwise AF demagnetisation to 90 mT to identify three components of magnetisation in banded chert carbonate exposed in a rail cut south of Blende Lake, 35 km northeast of Thunder Bay. There the beds show local folding and

brecciation of probable Keweenawan age (Moorhouse, 1960; Purucker, 1982, 1983). A pre-folding direction carried by magnetite ($D=89.5°$, $I=-77.4°$, $\alpha_{95}=3.4°$; pole position of 43.0°S,57.4°E) was interpreted by Purucker (1983) as earliest Keweenawan (1.2–1.15 Ga), or 1.65 or 2.1 Ga in age. A post-folding, Keweenawan direction was tentatively identified but not listed. Purucker (1983) regarded a younger post-folding direction carried by hematite ($D=320.0°$, $I=72.7°$, $\alpha_{95}=9.5°$; pole position of 64.7°N,218.6°E) as Cretaceous in age and related to the development of soft iron-ores in the Mesabi Range in Minnesota and the Gunflint Range in Ontario. Such ore now is common only in the Mesabi Range but Purucker (1983) speculated that Pleistocene glaciation may have removed similar ore in Ontario.

It is evident that previous palaeomagnetic studies of the Gunflint Formation and equivalent strata have produced a variety of seemingly inconsistent results and that further investigation of Gunflint palaeomagnetism is warranted.

3.2. Rove Formation

Books (1972) identified a direction of $D=240.9°$, $I=58.5°$ for five samples from the Rove Formation near Grand Portage, Minnesota. Pesonen (1979) obtained a similar direction of $D=254°$, $I=48°$ for one sample of unbaked Rove sediment from the Sibley Peninsula near Thunder Bay.

3.3. Sibley Group

DuBois (1962) provided results for 24 samples collected from the Sibley Group on the Sibley Peninsula and south of Lake Nipigon. He identified two, almost anti-parallel components with a combined mean direction of $D=78.0°$, $I=-51.5°$.

Robertson (1973a) studied the palaeomagnetism of the Sibley Group between Thunder Bay and Nipigon Bay. Stepwise thermal demagnetisation to 700 °C of one pilot specimen from each site was followed by bulk thermal demagnetisation of remaining specimens in three groupings at 480, 560 and 670 °C. Robertson (1973a) identified a normal direction ($D=240°$, $I=16°$) and a reverse direction ($D=48°$, $I=-5°$) that he regarded as early and which give a pole position of 20°S,214°E. An additional direction ($D=109°$, $I=-69°$) was assumed to have been acquired during intrusion of the Logan sills.

Pesonen (1979) obtained a direction of $D=225°$, $I=-2°$ for one sample of unbaked Sibley sediment, a result that he viewed as in reasonably good agreement

Table 1
Summary of natural remanent magnetisation (NRM) and anisotropy of magnetic susceptibility (AMS) of the Gunflint Formation and Sibley Group, Thunder Bay area, northwestern Ontario, Canada

k (μSI)	J_r (mA m^{-1})	Q	k_{max} (μSI)	k_{int} (μSI)	k_{min} (μSI)	A	L	F	t-factor
Sites 13–16: Gunflint Formation, Lower East Taconite Grouping #1									
420 ± 210	143 ± 72	3.91 ± 1.95	269	269	266	1.01	1.00	1.01	0.360
Sites 17–22: Gunflint Formation, Lower East Taconite Grouping #2									
255 ± 110	43.7 ± 17.8	4.47 ± 1.82	371	367	364	1.02	1.01	1.01	−0.462
Sites 05–12: Gunflint Formation, Upper Taconite									
14900 ± 5300	452 ± 160	0.76 ± 0.27	12500	12300	10900	1.14	1.02	1.12	0.724
Site 03: Gunflint Formation, Upper Chert Carbonate									
157 ± 13	1.25 ± 0.19	0.19 ± 0.05	(irregular shaped specimens prevented measurement of AMS)						
Site 23: Sibley Group conglomerate clasts									
207 ± 27	38.6 ± 5.03	7.57 ± 0.99	(randomly oriented clasts prevented calculation of tensor mean)						

Means ± standard error for k, low field magnetic susceptibility (μSI); J_r, intensity of natural remanent magnetisation (mA m^{-1}); Q, Königsberger ratio; k_{max}, maximum susceptibility from tensor means (μSI); k_{int}, intermediate susceptibility (μSI); k_{min}, minimum susceptibility (μSI); A, degree of anisotropy; L, linearity; F, foliation; t-factor, mean shape parameter $(2\ln k_{int} - \ln k_{max} - \ln k_{min})/(\ln k_{max} - \ln k_{min})$.

with the normal direction identified by Robertson (1973a).

4. Sampling of the Gunflint Formation and Sibley Group

4.1. Sampling sites

Eighty-two oriented samples were collected from 22 sites in the Gunflint Formation from quarries, river courses, and road and rail cuts within an area spanning 90 km between Nolalu to the southwest and Loon Lake to the northeast of Thunder Bay. Reconnaissance sampling of a variety of Gunflint facies provided material for a pilot palaeomagnetic study, and the most favourable rock types and sites were further sampled. Where possible, folds were sampled to provide fold tests, and clasts of Gunflint taconite were sampled from the basal conglomerate of the Sibley Group to permit a conglomerate test. Most sites were block-sampled, and others were sampled using a portable drill. Orientation was carried out using both sun and magnetic compasses. The mean of the local declination agreed within error with the published value of 1.3°E. The stratigraphic positions of sites in the Gunflint Formation are shown in Fig. 2 and site details are given in Appendix A.

4.2. Composition of Gunflint samples

Floran and Papike (1975) described the petrology of the relatively unmetamorphosed Gunflint Formation in Ontario and Minnesota. They identified two different types of Gunflint iron formation: (1) cherty iron-formation containing peloids and ooliths and represented by the lower and upper taconite and algal chert facies; (2) banded iron formation encompassing the chert carbonate and tuff argillite facies. Hematite, ranging from dispersed dust to massive granules, is common in the cherty rocks, whereas magnetite is sparsely distributed. Other constituent minerals include greenalite, minnesotaite, chamosite, stilpnomelane, calcite, ankerite and siderite.

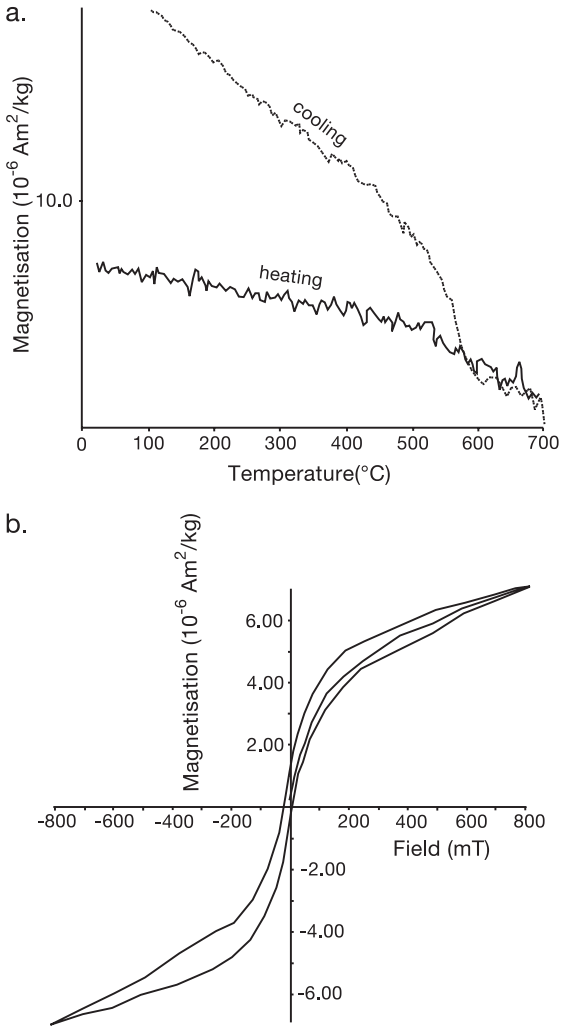
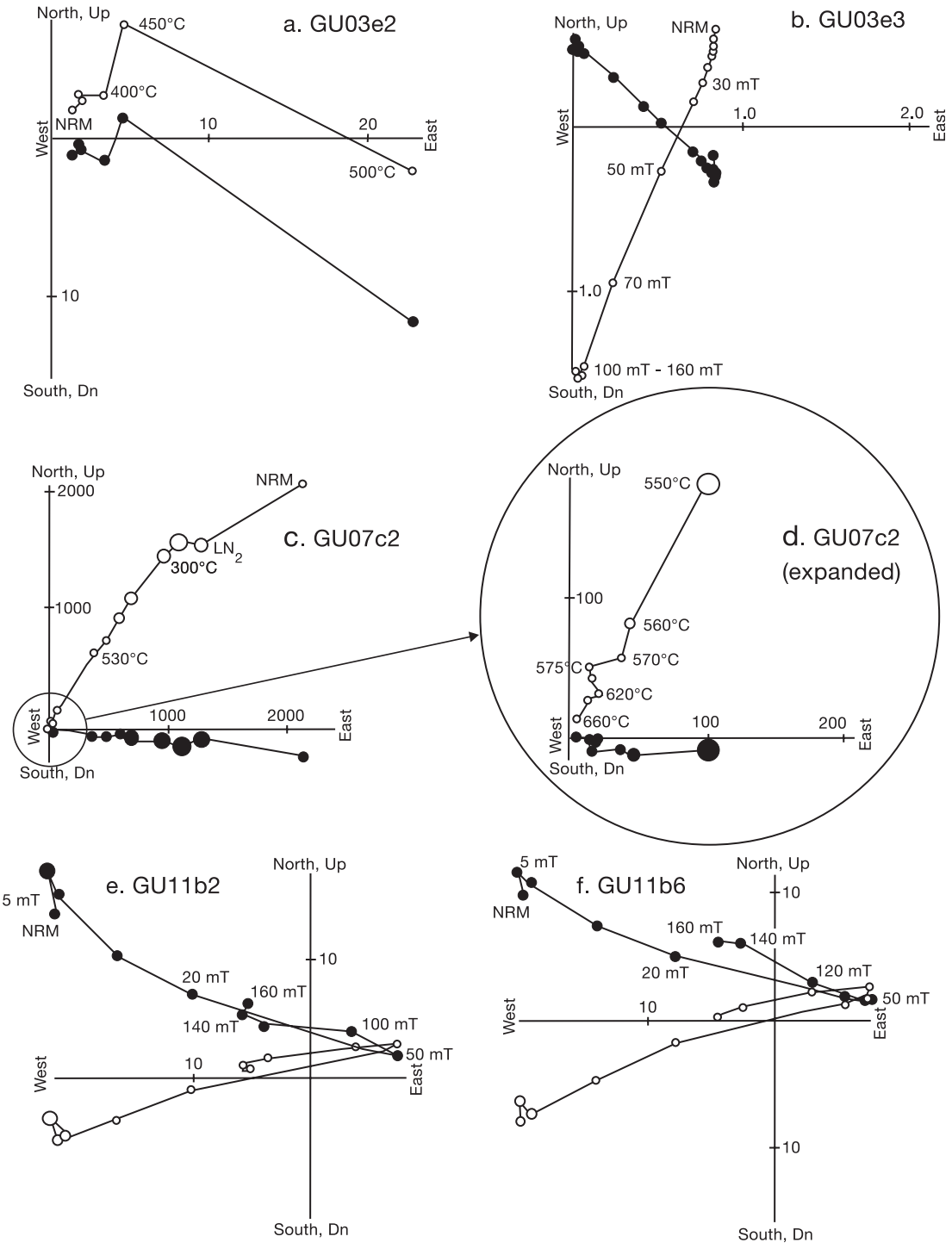


Fig. 4. (a) Saturation magnetisation versus temperature (J_s–T) curve and (b) hysteresis loop for sample GU12a. The heating curve shows the predominance of hematite, although magnetite clearly has been generated by heating as reflected in the cooling curve. The wasp-waisted loop is interpreted as indicating the presence of minor magnetite in the unheated sample. Sample number refers to site/sample.

Fig. 5. Orthogonal projections of remanence vectors after stepwise thermal demagnetisation (°C) and stepwise AF demagnetisation (mT). The LN_2 step after NRM refers to low-temperature demagnetisation using liquid nitrogen. Solid circles plot on the horizontal plane and the open circles on the vertical plane. The size of the circles reflects the angular standard deviation of the measurement. All units are mA m^{-1} (10^{-6} emu cc^{-1}). Specimen numbers refer to site/sample/specimen. See text for details.


a. GU03e2
North, Up
450°C
400°C
NRM
10
20
West
East
500°C
10
South, Dn
b. GU03e3
North, Up
NRM
30 mT
1.0
2.0
West
East
50 mT
1.0
70 mT
100 mT - 160 mT
South, Dn
c. GU07c2
North, Up
2000
NRM
LN2
300°C
1000
530°C
1000
2000
West
East
South, Dn
d. GU07c2
(expanded)
North, Up
550°C
100
560°C
575°C
570°C
620°C
660°C
100
200
West
East
South, Dn
e. GU11b2
5 mT
NRM
North, Up
10
20 mT
160 mT
140 mT
100 mT
50 mT
10
West
East
South, Dn
f. GU11b6
5 mT
NRM
North, Up
10
160 mT
140 mT
20 mT
120 mT
50 mT
10
West
East
10
South, Dn

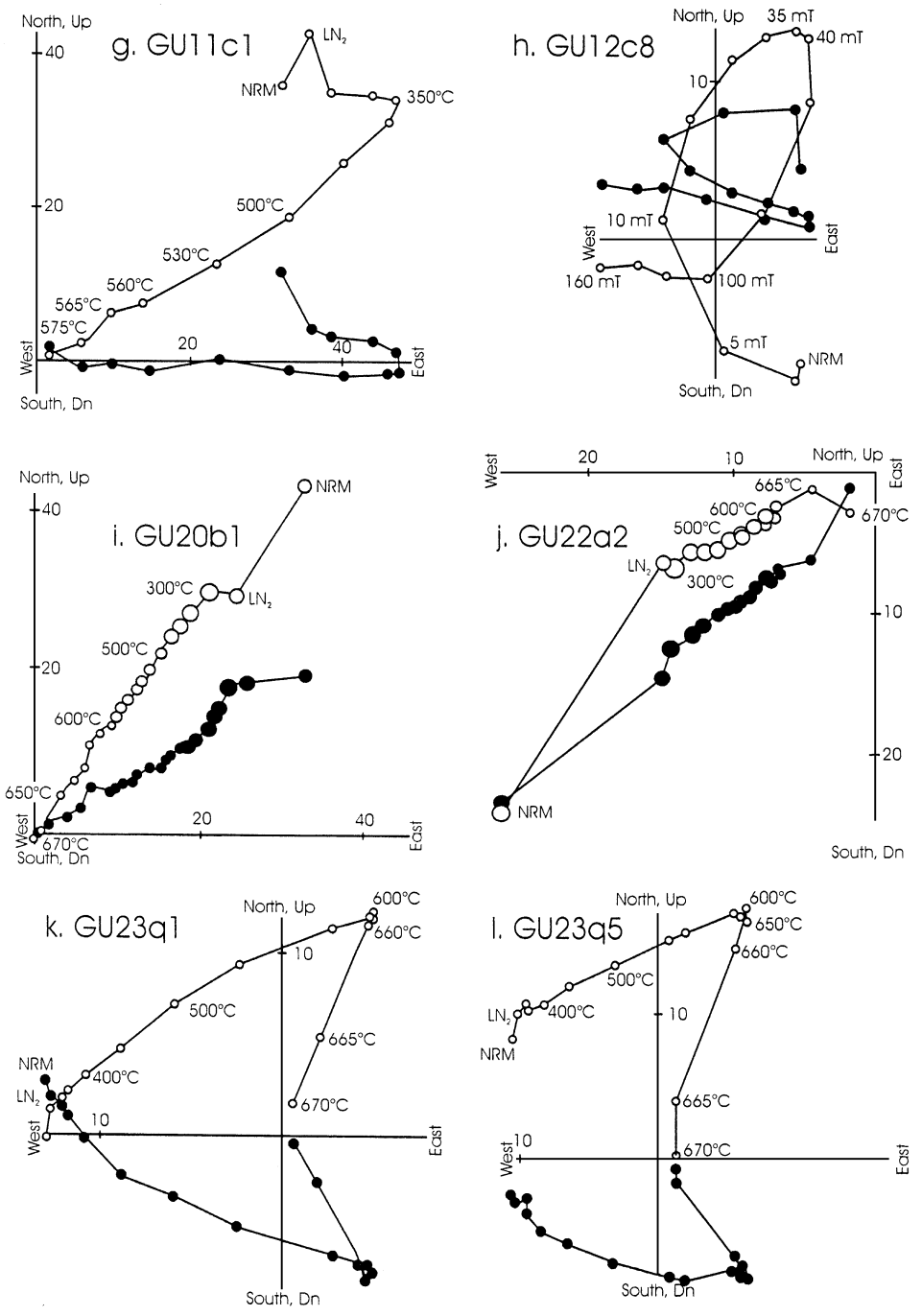


Fig. 5 (*continued*).

The minerals identified by X-ray diffraction analysis in this study and the total Fe content (expressed as Fe_2O_3) for samples of the Gunflint Formation from the principal sampling sites are as follows:

- Sites 02, 16, 20, 22, in grainstones of the Lower East Taconite. Dolomite, hematite, quartz, magnetite, calcite (trace), chlorite (trace). Mean Fe_2O_3T = 11.98%, range = 6.56–23.68% (6 samples).
- Site 17, concentration of specularite in the Lower East Taconite. Hematite, quartz (minor). Fe_2O_3T = 79.11% (1 sample).
- Sites 08, 10, 11, in peloidal facies of the Upper Taconite. Quartz, siderite, calcite, hematite, stilpnomelane, greenalite, dolomite (trace), chlorite (trace). Mean Fe_2O_3T = 26.62%, range = 17.92–42.28% (4 samples).
- Sites 03, 04, in the Upper Chert Carbonate. Quartz, siderite, dolomite, hematite (trace), chlorite (trace).

The iron-rich rocks containing >15% Fe of primary origin (depositional or diagenetic) are iron formation as defined by James (1966).

5. Rock magnetism

5.1. Laboratory techniques

Bulk susceptibility (k) was measured with a transformer bridge with a sensitivity of 10^{-7} SI and calibrated using paramagnetic salts (Ridley and Brown, 1980). Anisotropy of magnetic susceptibility (AMS) was measured using a Digico anisotropy delineator. The operating system automatically corrects results for the delineator calibration error noted by Veitch et al. (1983) and Hrouda et al. (1983). Thermomagnetic curves and susceptibility versus temperature measurements were obtained using a Petersen variable-field translation balance that operates as a harmonic oscillator with damping in the forced oscillation mode. Fast Fourier transform of the signal enhances its sensitivity by orders of magnitude over conventional magnetic balances. The variable-field translation balance can easily detect diamagnetic materials. A furnace/cooling unit allows samples to be measured between − 175 and 700 °C. Samples are usually small cylindrical plugs (5 mm diameter × 5 mm high) although crushed samples (~ 200 mg) also can be used if packed tightly enough. The small plugs were cored from off-cuts of standard-sized palaeomagnetic specimens.

5.2. Results

Bulk susceptibility, natural remanent magnetisation (NRM) and Königsberger ratio (Q) determinations on 315 specimens of the Gunflint Formation fall into four broad groupings. The Sibley conglomerate clasts form a fifth grouping. The group means and their standard errors are given in Table 1.

Data for the Sibley conglomerate taconite clasts and the Lower East Taconite groupings are similar, particularly the unusually high Q values of ~ 4–8. This finding is consistent with the clasts having been derived from the Gunflint Formation. The highest bulk susceptibilities were observed in the Upper Taconite, which has a mean k of ~ 0.015 SI. Using the relationship of Puranen (1989), $k = 3.47f$ SI where f is the volume fraction of magnetite, the susceptibility of the Upper Taconite indicates 0.4% magnetite content by volume. This estimate ignores the contribution from hematite which, if taken into account, would lower the estimate only slightly because hematite has a very much lower susceptibility than has magnetite. The Lower East Taconite appears to contain even less magnetite ($\ll$0.1%).

AMS measurements for the Gunflint Formation groupings also are shown in Table 1. Minimum susceptibility axes are clustered near the bedding poles, indicating that the Gunflint Formation has not undergone dynamic metamorphism. The anisotropy tensor mean for the Sibley conglomerate taconite clasts is not meaningful because the original orientation of the clasts cannot be determined. The AMS of the Upper Chert Carbonate was not measured because these specimens were not of standard size and shape.

Saturation magnetisation versus temperature (J_s–T) and hysteresis loop measurements (Fig. 4) were conducted using the variable-field translation balance. The J_s–T results for sample GU12a are typical for the taconite and indicate that the main magnetic mineral is hematite, with a Curie temperature of ~ 680 °C. However, after heating there is clearly a phase with a lower Curie temperature around

Table 2
Summary of palaeomagnetic results for the present study of the Gunflint Formation and Sibley Group, Thunder Bay area, northwestern Ontario, Canada

Max. T_{ub} (mT)	Polarity	n	D_h (°)	I_h (°)	K	α_{95} (°)	ξ_1	D_b (°)	I_b (°)	K	α_{95} (°)	ξ_1	0.05 critical value for fold test
Sites 02, 06–22: Gunflint Formation, quarries in Upper Taconite, Thunder Bay, and roadcut in Lower East Taconite, Loon Lake													
580 °C	N	11	298.6	44.1	55.4022	6.2	0.833	293.9	47.3	38.1803	7.5	0.309	3.865 Inconclusive
580 °C	R	29	86.7	−54.6	21.9057	5.8	5.233	88.8	−55.3	15.3013	7.1	5.095	6.264 Inconclusive
50 mT	N	10	289.2	15.9	66.3908	6.0	3.278	287.8	18.0	66.3917	6.0	3.278	3.685 Inconclusive
Mt	N	21	293.4	30.8	20.2560	7.2	0.438	290.6	33.7	17.9466	7.7	0.318	5.335 Inconclusive
Mt	R	29	86.7	−54.6	21.9057	5.8	1.844	85.4	−54.7	14.2283	7.4	13.197	6.264 Strong negative
Pole, Mt Normal (GMN): Lat. = 27.7°N, Long. = 175.0°E (dp = 4.4°, dm = 8.0°).													
Pole, Mt Reverse (GMR): Lat. = 23.5°N, Long. = 208.2°E (dp = 5.7°, dm = 8.1°).													
680 °C	N	11	243.6	23.6	58.6252	6.0	3.054	248.2	4.4	6.0792	20.2	7.343	3.865 Negative
680 °C	R	60	64.3	−52.1	31.4051	3.3	6.327	74.2	−45.2	9.0010	6.5	42.927	9.010 Strong negative
160 mT	R	19	88.1	−47.2	26.9001	6.6	0.439	82.6	−47.3	26.9001	6.6	0.439	5.075 Nil[a]
Ht	N	11	243.6	23.6	58.6252	6.0	3.054	248.2	4.4	6.0792	20.2	7.343	3.865 Negative
Ht	R	79	70.3	−51.4	25.2010	3.2	23.583	71.6	−50.4	25.3245	3.2	0.138	10.339 Syn-folding[b]
Pole, Ht Normal (GHN): Lat. = 6.6°S, Long. = 208.6°E (dp = 3.4°, dm = 6.3°).													
Pole, Ht Reverse (GHR): Lat. = 11.9°N, Long. = 215.0°E (dp = 2.8°, dm = 4.2°).													
Site 03: Gunflint Formation, railcut in Upper Chert Carbonate south of Blende Lake													
Mt	R	18	158.1	−70.0	9.5365	11.8	16.390	97.3	−78.2	31.5505	6.3	3.670	4.940 Strong positive[c]
Ht	N	20	1.2	76.7	92.7373	3.4	4.140	159.7	86.9	10.5098	10.6	18.109	5.207 Strong negative[c]
Pole, Mt[d] Reverse (BMR): Lat. = 46.7°N, Long. = 237.0°E (dp = 11.1°, dm = 11.8°).													
Pole, Ht Normal (BHN): Lat. = 74.2°N, Long. = 272.8°E (dp = 5.6°, dm = 6.3°).													
Site 23: Sibley Group, clasts of Gunflint taconite in basal conglomerate, Pass Lake													
Mt	N	20	282.7	33.4	19.4807	7.6	0.210	283.2	36.3	19.4807	7.6	0.210	5.207 Nil[a]
Ht	N	13	254.5	56.2	25.6055	8.4	1.306	253.2	59.1	25.6055	8.4	1.306	4.200 Nil[a]
Ht	R	11	110.6	−55.7	31.2930	8.3	0.965	112.2	−58.5	31.2930	8.3	0.965	3.865 Nil[a]
Pole, Mt Normal (SMN): Lat. = 21.8°N, Long. = 184.3°E (dp = 4.9°, dm = 8.6°).													
Pole, Ht Normal (SHN): Lat. = 17.7°N, Long. = 217.0°E (dp = 8.7°, dm = 12.0°).													
Pole, Ht Reverse (SHR): Lat. = 39.0°N, Long. = 194.6°E (dp = 8.5°, dm = 11.8°).													

T_{ub}, unblocking temperature; n, number of specimens; D_h, I_h, mean declination and mean inclination with respect to present horizontal; D_b, I_b, mean declination and mean inclination with respect to bedding; R, resultant length of n unit vectors; K, precision parameter; α_{95}, semi-angle of cone of 95% confidence about mean direction; ξ_1 is the fold test statistic of McFadden (1990); dp, dm, semi-minor and semi-major axes of polar error ellipse.

For sites 02, 06–22 and 03, data for $T_{ub} \leq 580$ °C and $H_{cr} \leq 50$–100 mT are combined to give overall magnetite results (Mt), and data for 580 °C $\leq T_{ub} \leq 680$ °C and 100 mT $\leq H_{cr} \leq 160$ mT are combined to give overall hematite results (Ht).

[a] Nil: no test (only one limb of fold available/sampled).

[b] Minimum ξ_1 at 20% unfolding.

[c] ξ_2 convention used.

[d] Pole calculated in stratigraphic coordinates (sole positive fold test).

575 °C that we believe to be magnetite. This is discussed further in Section 6.2. The hysteresis loop for GU12a also is typical for the taconite and shows a wasp-waisted loop that is interpreted to reflect a minor amount of magnetite. This situation is not common in natural samples because it requires the occurrence of hematite and magnetite in just the right proportions (~100:1) such that they contribute approximately equally to the magnetisation (Dunlop and Özdemir, 1997).

6. Palaeomagnetism

6.1. Laboratory techniques

Routine palaeomagnetic laboratory methods (Collinson, 1983; Butler, 1992) were employed. Remanent magnetisations were measured using a 2G 755R three-axis cryogenic magnetometer interfaced to a computer. The demagnetisers used were an in-line 2G 600 series AF demagnetiser and the CSIRO automated three-stage carousel furnace that is housed in a 4-m 10-coil Helmholtz set with automatic feedback maintaining zero-field (<5 nT). Following low-temperature pre-treatment (Schmidt, 1993), all samples were subjected to stepwise thermal demagnetisation. In addition, some samples were subjected to stepwise AF demagnetisation and AF pre-treatment prior to measuring after high-temperature thermal demagnetisation steps.

Components of magnetisation were isolated using an interactive version of Linefind (Kent et al., 1983), whereby linear segments are fitted to data points weighted according to the inverse of their measured variances. Because some specimens carry remanence of both normal and reverse polarities, indicating that remanence acquisition has spanned at least one geo-

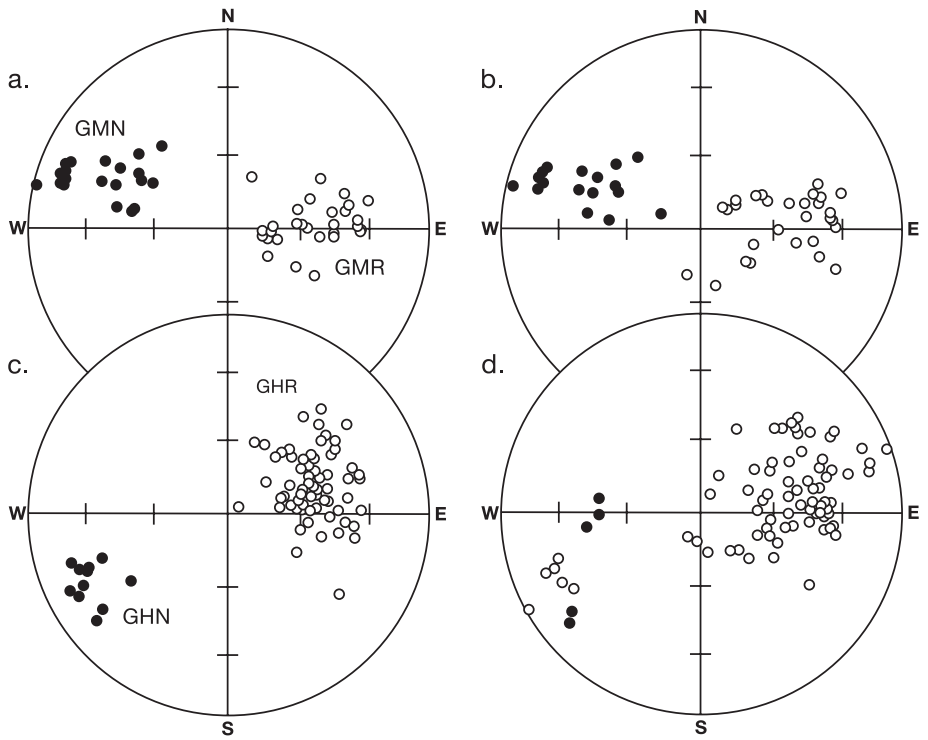


Fig. 6. Stereographic projections of components isolated in Gunflint Formation taconite (sites 02, 06–22) by stepwise thermal and stepwise AF demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere. (a) and (b) are components carried by magnetite, before and after bedding-correction, respectively. (c) and (d) are components carried by hematite, before and after bedding-correction, respectively.

magnetic reversal, acquisition has occurred over sufficient time to adequately average secular variation. Accordingly, unit weight was assigned to specimens in the statistical analysis. Excluding premature directional instability evidently caused by the oxidation of siderite in some specimens, the highest temperature to which any particular component remains stable, i.e. the highest unblocking temperature observed, is generally taken as an indication of the Curie temperature of the mineral carrying that component.

6.2. Results

Typical demagnetisation behaviour for various lithologies is shown in Fig. 5 and directions determined from line fitting are plotted as stereo projections in Figs. 6–8. Palaeomagnetic results for the present study are summarised in Table 2.

Numerous specimens, mostly from sites 03, 04, 06 and 08–12, acquired spurious components of magnetisation while cooling after heat treatment at 300–500 °C (Fig. 5a). Such unstable magnetisation is ascribable to the presence of siderite, which oxidises to magnetite on heating (Pan et al., 2000). X-ray diffraction analysis confirmed the presence of siderite in samples of the Gunflint Formation that displayed this instability.

Other specimens from the Gunflint Formation commonly yielded two components of remanence. Thermal demagnetisation generally defined the components satisfactorily for the taconite samples (sites 02, 06–22) (Fig. 5c,d,g), although AF demagnetisation gave better definition of the components for some samples (sites 11 and 12). Specimens GU11b2, GU11b6 and GU12c8 (Fig. 5e,f,h) each show two components in sharp opposition. Between 2.5 and 50 mT, a westerly down component was removed, followed by an easterly up component to 160 mT. A third component remained with coercivity of remanence (H_{cr})>160 mT and could not be thermally

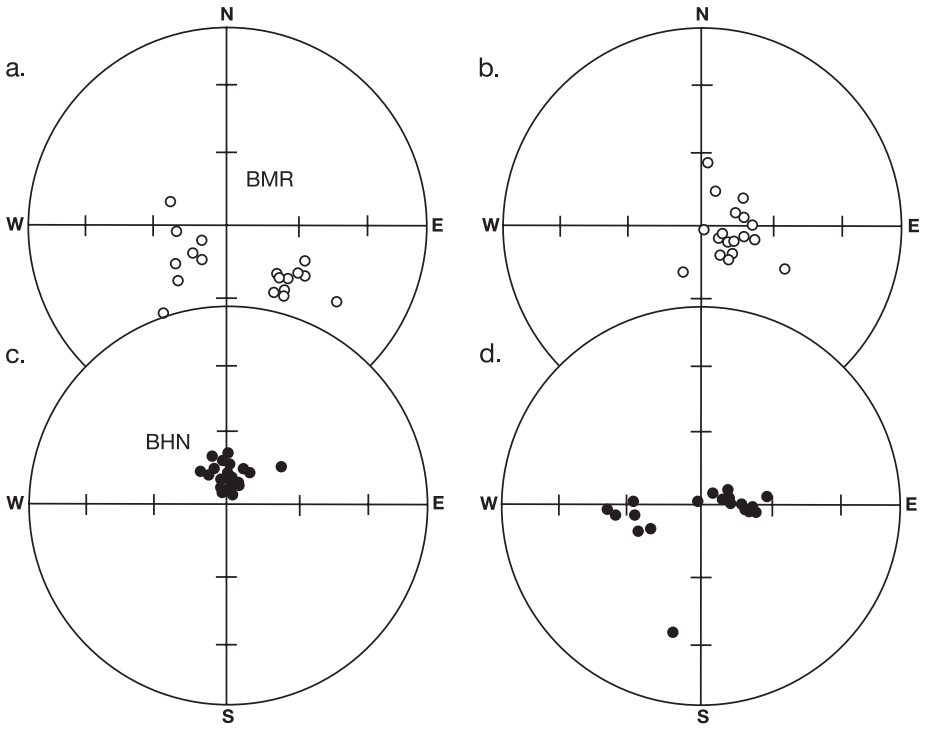


Fig. 7. Stereographic projections of components isolated in Gunflint Formation chert carbonate (site 03) by stepwise AF demagnetisation only. These samples could not be thermally demagnetised because of thermo-chemical alteration. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere. (a) and (b) are components carried by magnetite, before and after bedding-correction, respectively. (c) and (d) are components carried by hematite, before and after bedding-correction, respectively.

demagnetised because of the thermo-chemical alteration discussed above. Fig. 5i and j shows good examples of thermal demagnetisation of normal and reverse components carried by hematite.

The normal and reverse directions in their in situ and bedding dip-corrected coordinates for all taconite sites (02, 06–22) are plotted both for the magnetite and hematite components in Fig. 6. There is a general tendency for the groupings to scatter on bedding-correction. Table 2 gives the results of applying the fold test of McFadden (1990) to the groupings. Most fold-test results for these sites range from inconclusive to strongly negative, implying that these magnetisations were acquired after folding. Evidently, the Gunflint Formation in our sampling area has undergone widespread remagnetisation. One set of directions, the reverse hematite, yielded an apparent syn-folding result with a minimum ξ_1 at 20% unfolding (where ξ_1 is the fold test statistic of McFadden, 1990). It is uncertain if this really indicates acquisition of magnetisation during folding or whether it is just a Type-II statistical error, i.e. a situation where a statistic erroneously rejects the null hypothesis. The null hypothesis is that there is no correlation between dip direction and displacement of sample directions from the mean direction in geographical coordinates. Given the predominance of post-folding remanences present in the Gunflint Formation at sites 02, 06–22 it is perhaps more conservative to assume the latter.

Specimens of chert carbonate from site 03, e.g. GU03e2, displayed marked increase in intensity with each heating step above 350 °C (Fig. 5a). The magnetisation increased from ~ 1 to >20 mA m^{-1}, at which point thermal demagnetisation was abandoned for such specimens. Comparison with AF demagnetisation behaviour of companion specimen GU03e3 is startling. From 2.5 to 160 mT, a single component is demagnetised, leaving a hard stable

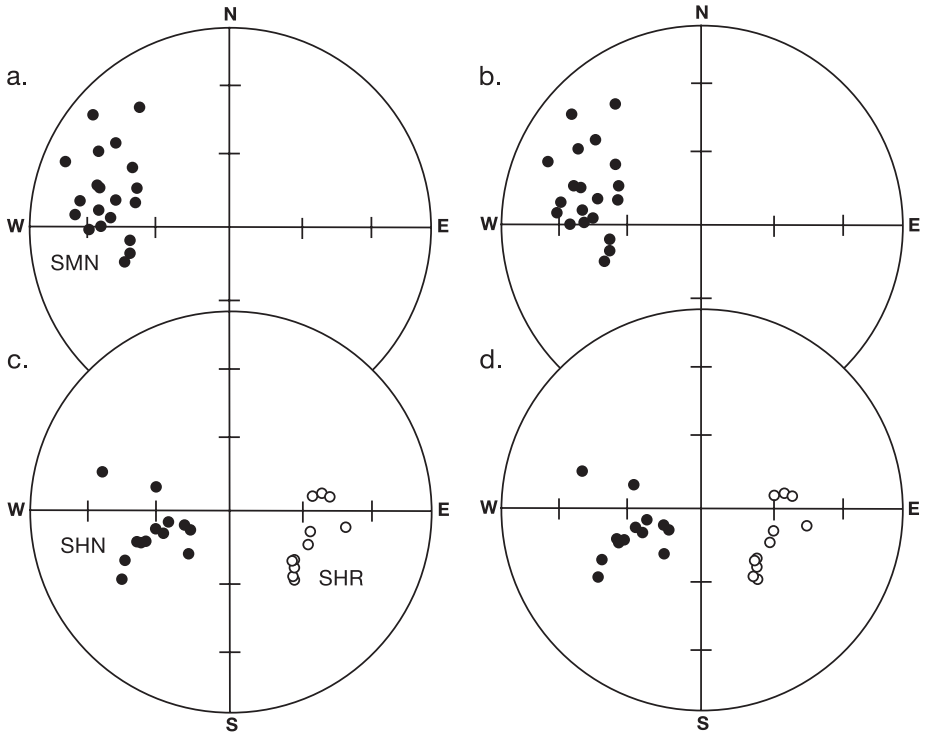


Fig. 8. Stereographic projections of components isolated in Gunflint Formation taconite clasts from the Sibley Group basal conglomerate (site 23) by stepwise thermal and stepwise AF demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere. (a) and (b) are components carried by magnetite, before and after bedding-correction, respectively. (c) and (d) are components carried by hematite, before and after bedding-correction, respectively.

component with H_{cr}>160 mT (Fig. 5b). The carrier of the latter component is assumed to be hematite, whereas the carrier of the lower coercivity component probably is magnetite. Because these samples reacted adversely to thermal demagnetisation, we resorted to the 'stable end-point' method for identifying the direction of the hematite component. The reverse magnetite directions, both in situ and bedding dip-corrected, are plotted in Fig. 7a and b, respectively. The magnetite component passes the fold test and is considered to be pre-folding, whereas the hematite component (Fig. 7a and d) fails the fold test and therefore is post-folding.

The Sibley conglomerate taconite clasts, which were derived from the Gunflint Formation, also revealed two components, a westerly down normal component and an easterly up reverse component (Fig. 5k and l). Based on Curie temperatures, the normal component evidently is carried both by magnetite and hematite, whereas the reverse component is carried only by hematite. These directions are plotted in their in situ and bedding dip-corrected coordinates in Fig. 8. The clustering of these directions clearly indicates a negative conglomerate test, which is consistent with the general result of the fold tests that indicate the Gunflint Formation has been widely remagnetised.

7. Discussion

7.1. Taconite

The taconite facies of the Gunflint Formation possesses both normal and reverse components, each carried both by magnetite and hematite. However, there is a systematic difference in the axes of the magnetite and the hematite mean directions, the hematite declinations being displaced anticlockwise with respect to those of the magnetite components. The distinctive declinations are present in sites containing only hematite as well as in those containing both magnetite and hematite, so there would not appear to be some peculiar interaction causing the displacement. Furthermore, as is displayed by the Logan intrusions and many other Keweenawan igneous rocks, the mean inclination of the taconite normal directions is shallower than that of the reverse directions (Table 3). The Gunflint hematite and magnetite components both fail the reversal test of McFadden and McElhinny (1990). The critical angle for the magnetite component is 9.0° whereas the observed angle is 30.5°, and the critical angle for the hematite component is 6.9° whereas the observed angle is 28.3° (Table 4). While the greater part of these differences is ascribable to inclination, a substantial amount is due to declination. The implication of a substantial difference in declination is discussed in Section 7.6.

Four separate pole positions are calculated for taconite sites 02 and 06–22 (Table 2). Each of these poles (GMN, GMR, GHN, GHR) plots close to the Logan Loop (see DuBois, 1962; Robertson and Fahrig, 1971) in the apparent polar wander (APW) path shown in Fig. 9 and their polarities are consistent with those of other Logan Loop poles, notwithstanding exceptions as noted by Symons et al. (1994). Table 3 lists representative Keweenawan and Sibley Group palaeomagnetic directions under their polarities to compare with overprint directions we identified for the Gunflint Formation and Sibley Group, Thunder Bay area.

The positions of the GHN and GMR poles (both post-folding) and the GHR pole on the APW path (Fig. 9) suggest that the folding in the Lower East Taconite (sites 02, 13–22) pre-dates the Sibley Group. The origin of this folding is not apparent, but our results suggest it is older than the folding and intense brecciation of apparent Keweenawan age displayed by the Upper Chert Carbonate at site 03 (see Section 7.2).

7.2. Upper Chert Carbonate

The palaeomagnetic pole positions determined for the Upper Chert Carbonate (site 03) are given in Table 2. The BMR pole, which has reverse polarity, falls on the apex of the Logan Loop (Fig. 9). As has been noted by many previous workers, the polarities of poles close to the apex of the Logan Loop are invariably reverse. The BMR component is pre-folding (Table 2), which is consistent with the folding and brecciation of the chert carbonate at site 03 having occurred during intrusion of the middle Keweenawan Logan sills (see Section 2).

Both the magnetite and hematite components were identified previously by Purucker (1983), who also

Table 3
Representative palaeomagnetic directions (post-1969) for Keweenawan igneous rocks, Sibley Group and Rove Formation

D (°)	*I* (°)	Rock unit	Locality	Reference
Normal directions				
Keweenawan igneous rocks				
Dykes				
298/290	27/28	Dykes	Thunder Bay area	Pesonen (1979)
Sills				
277	47	Logan diabase intrusives	Duluth, Minnesota	Beck (1970)
283	57	Lester River sill	Duluth, Minnesota	Beck (1970)
286	49	Endion sill	Duluth, Minnesota	Beck (1970)
Volcanic rocks				
287	47	North Shore Volcanics	N. Michigan	Palmer (1970)
293	35	Mamainse Point Lavas	E. Lake Superior	Robertson (1973b)
296	39	Osler Volcanics (upper)	Nipigon Strait, Ontario	Halls (1974)
299	37	Mamainse Point Lavas	E. Lake Superior	Halls and Palmer (1981)
Gabbro				
280	56	Duluth Gabbro	Duluth, Minnesota	Beck (1970)
290	53	Layered series	Duluth, Minnesota	Beck (1970)
294	56	Anorthositic gabbro	Duluth, Minnesota	Beck (1970)
Sibley Group				
240	16	Sibley Group	Sibley Peninsula and Nipigon area	Robertson (1973a)
252	20	Sibley Group (unbaked)	Thunder Bay area	Pesonen (1979)
291	31	Baked rocks, dykes	Thunder Bay area	Pesonen (1979)
Rove Formation				
240	58	Rove Formation	Grand Portage, Minnesota	Books (1972)
254	48	Rove Formation	Thunder Bay area	Pesonen (1979)
Reverse directions				
Keweenawan igneous rocks				
Dykes				
108	− 78	Baraga dykes (AF demag.)	N. Michigan	Pesonen and Halls (1979)
134	− 81	Baraga dykes (thermal)	N. Michigan	Pesonen and Halls (1979)
115	− 68	Marquette dykes (AF)	N. Michigan	Pesonen and Halls (1979)
107	− 68	Marquette dykes (thermal)	N. Michigan	Pesonen and Halls (1979)
123	− 73	Dykes	Thunder Bay area	Pesonen (1979)
Sills				
110	− 69	Logan sills	Thunder Bay area	Palmer (1970)
108	− 73	Logan sills	Lake Nipigon area, Ontario	Palmer (1970)
100	− 72	Logan sills	Thunder Bay area	Robertson and Fahrig (1971)
105	− 68	Sills	Thunder Bay area	Pesonen (1979)
Volcanic rocks				
116	− 62	North Shore Volcanics	N. Michigan	Palmer (1970)
117	− 64	Osler Volcanics	Lake Nipigon–Black Bay, Ontario	Palmer (1970)
106	− 74	Basalt	Alona Bay, E. Lake Superior	Palmer (1970)
104	− 74	Mamainse Point Lavas	E. Lake Superior	Robertson (1973b)
115	− 75	Osler Volcanics (lower)	Nipigon Strait, Ontario	Halls (1974)
Gabbro				
91	− 65	Gabbro, Logan intrusives	Duluth, Minnesota	Beck (1970)
Sibley Group				
48	− 5	Sibley Group	Sibley Peninsula and Nipigon area	Robertson (1973a)
225	− 2	Sibley Group	Thunder Bay area	Pesonen (1979)
147	− 68	Sibley Group (baked)	Sibley Peninsula and Nipigon area	Robertson (1973a)
116	− 68	Sibley Group (baked)	Thunder Bay area	Pesonen (1979)

Further data are given by Palmer (1970), Robertson and Fahrig (1971), Pesonen and Halls (1979), Pesonen and Nevanlinna (1981), Halls and Pesonen (1982), and in Table 4.

Table 4
Reversal tests for directions for the Gunflint Formation and Sibley Group conglomerate (present study), Mamainse Point Lavas (Palmer, 1970; Robertson, 1973b) and alkali syenite–carbonatite complexes in the Kapsukasing Structural Zone, Ontario (Costanzo-Alvarez et al., 1993; Symons et al., 1994)

Unit	D (°)	I (°)	N	R	K	α_{95} (°)	D (°)	I (°)	N	R	K	α_{95} (°)	γ_o (°)	γ_c (°)	Reversal test
	Normal						Reverse								
Gunflint Mt	293.4	30.8	21	20.013	20.256	7.24	86.7	−54.6	29	27.722	21.906	5.85	30.5	9.0	Negative
Gunflint Ht	243.6	23.6	11	10.829	58.625	6.01	70.3	−51.4	79	75.905	25.201	3.23	28.3	6.9	Negative
Sibley Ht	254.5	56.2	13	12.531	25.606	8.35	110.6	−55.7	11	10.680	31.293	8.29	20.0	11.4	Negative
Mam N1-R1	303.1	39.7	42	40.230	23.152	4.68	104.2	−57.4	7	6.862	43.312	9.28	21.5	10.1	Negative
Mam N2-R1	290.2	40.4	20	19.258	25.58	6.58	104.2	−57.4	7	6.862	43.312	9.28	17.4	10.9	Negative
Mam N2-R2	290.2	40.4	20	19.258	25.58	6.58	125.9	−73.7	25	23.942	22.68	6.21	34.1	9.0	Negative
Chipman Lake	298.9	50.5	12	11.602	27.610	8.41	103.8	−37.2	2	1.952	20.833	57.81	17.1	39.2	Indeterminate
Seabrook Lake	313.0	47.9	10	9.255	12.084	14.48	116.5	−42.8	3	2.746	7.874	47.25	12.6	42.0	Indeterminate
Nemegosenda[a]	304.0	51.2	7	6.801	30.151	11.17	112.1	−46.1	9	8.398	13.285	14.66	9.4	17.9	Positive C
Shenango[a]	312.4	54.5	7	6.655	17.371	14.90	123.9	−57.1	4	3.765	12.766	26.75	5.5	27.2	Indeterminate
Lackner Lake and Borden	309.1	50.5	3	2.993	307.692	7.04	108.7	−34.6	2	1.963	27.027	50.07	21.7	39.8	Indeterminate
Combined (above five units)	306.3	50.7	39	37.200	21.111	5.11	113.0	−45.7	20	18.644	14.012	9.04	10.2	10.3	Positive C

D, declination; I, inclination; N, number of units; R, resultant of unit vectors; K, precision parameter; α_{95}, half-angle of 95% cone-of-confidence; γ_o, observed angular difference between axes of normal and reverse; γ_c, critical angular difference between axes of normal and reverse.

[a] Minor errors in the calculation of K and α_{95} are apparent in Costanzo-Alvarez et al. (1993, their Tables 2 and 3) and were corrected prior to executing these reversal tests.

opted for AF demagnetisation (see Section 3.1). Although he did not explain why thermal demagnetisation was not used, we assume it was related to the thermo-chemical alteration of siderite (see Section 6.2). Purucker (1983) assigned the pre-folding component to the earliest Keweenawan or perhaps older. However, the reference pole path for North America has changed somewhat with the introduction of the key palaeopole approach (Buchan et al., 2001), making a pre-Keweenawan age untenable. Purucker (1983) assigned the post-folding component to the Cretaceous.

As discussed in Section 3.1, Symons (1966) found a mean direction of $D=192°$, $I=79°$ for chert carbonate of the Gunflint Formation in the Thunder Bay area, with a pole position of 28°N,266°E. This result differs from any found in our study, although the steep down inclination is reminiscent of the post-folding hematite component identified here and by Purucker (1983) for chert carbonate from site 03. However, no field test exists to constrain the data of Symons (1966). Likewise, we found no direction similar to directions reported by Atuanya (1982) which he interpreted as overprints related to the Hudsonian Orogeny. Both the studies by Symons (1966) and Atuanya (1982) employed 'bulk cleaning' techniques and the directions thus identified may be biased by unresolved components. We have calculated a 'bulk cleaned' 70 mT equivalent from our data for site 03 and found a mean direction of $D=353°$, $I=79°$, which is still quite different from that found by Symons (1966). We suggest that improvements in AF cleaning technology may explain the difference.

7.3. Sibley Group conglomerate

The palaeomagnetic pole positions for taconite clasts from the basal conglomerate of the Sibley Group (site 23) are given in Table 2. The magnetite pole (SMN) plots on the descending limb of the Logan Loop (Fig. 9) quite near the Gunflint magnetite normal polarity pole (GMN). The normal and reverse hematite poles (SHN, SHR) plot on opposite sides of the loop apex near 1.2 and 1.1 Ga, respectively.

The results for the Sibley conglomerate clearly indicate that the clasts have been remagnetised. The rock magnetic properties of the clasts are similar to those of the Gunflint taconite, supporting other evidence that the clasts were derived from this facies. In addition, the components carried by the

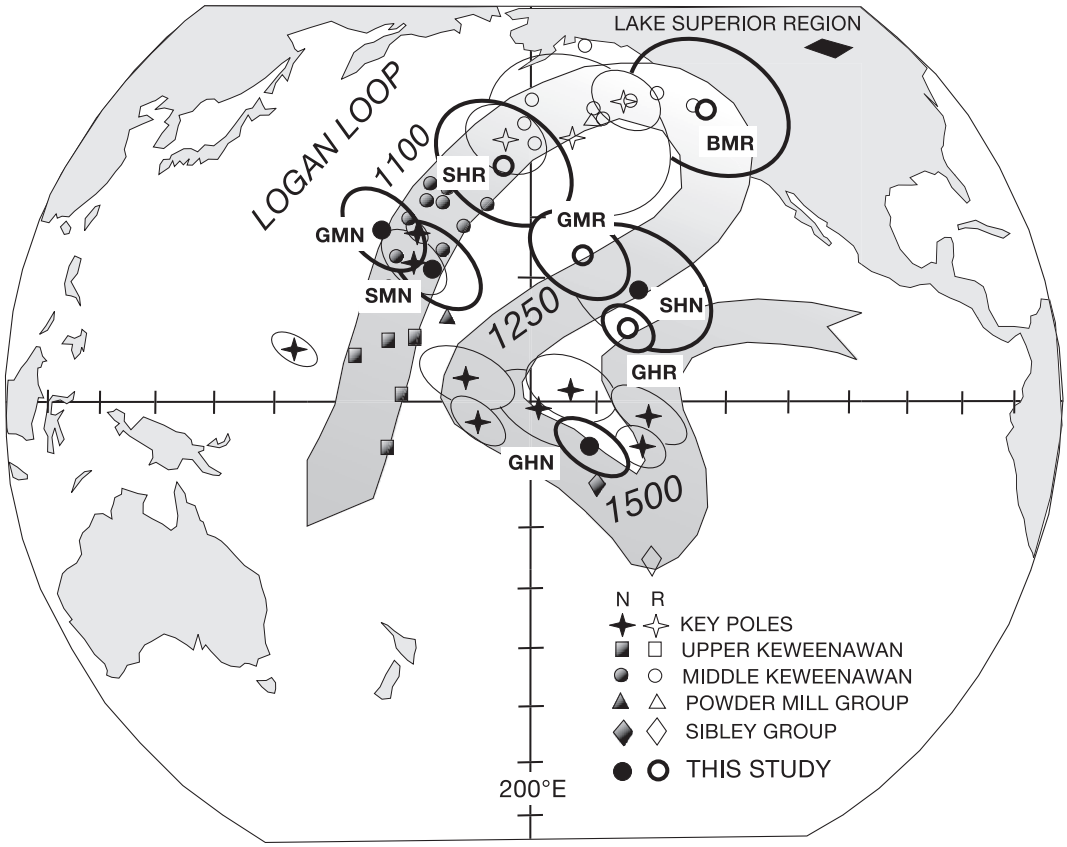


Fig. 9. Palaeomagnetic poles for the Gunflint Formation determined by the present study and palaeomagnetic poles for Keweenawan igneous and sedimentary rocks in the Lake Superior region. Adapted from Pesonen and Halls (1979) with key poles added from Buchan et al. (2001). GMN, Gunflint magnetite normal; GMR, Gunflint magnetite reverse; GHN, Gunflint hematite normal; GHR, Gunflint hematite reverse; BMR, Blende Lake (Gunflint) magnetite reverse; SMN, Sibley (conglomerate) magnetite normal; SHN, Sibley (conglomerate) hematite normal; SHR, Sibley (conglomerate) hematite reverse.

clasts are similar to those in the taconite. There seems little doubt that the clasts and the Gunflint Formation were remagnetised sometime after the Sibley Group was deposited. Apparently, the Sibley Group itself has not been overprinted, indicating a particular vulnerability to remagnetisation of the Gunflint-derived clasts.

The hematite component fails the reversal test of McFadden and McElhinny (1990), with the critical angle being 11.4° and the observed angle 20.0° (Table 4). As shown by the Gunflint taconite, not all of the asymmetry resides in the inclination. Indeed, for the Sibley conglomerate clasts almost all the asymmetry is in the declination. The importance of this finding for the interpretation of the asymmetries observed in the present study is discussed in Section 7.6.

7.4. Origin of overprinting

Some of the directions we identified in the Gunflint Formation and Sibley Group conglomerate are close to those in 1.1-Ga Keweenawan igneous rocks and older rocks in the Lake Superior region (Table 3). At a glance, GMN is similar to Copper Harbor directions, both GHN and GHR are similar to Sibley Group directions (as found by DuBois, 1962, rather than those of Robertson, 1973a), and Sibley conglomerate components mimic those for the Duluth Complex, Rove Formation and Logan sills. The BMR pre-

folding component for the chert carbonate also is similar to Logan sill directions.

The apparent asymmetrical reversals in our data also are reminiscent of those commonly observed in Keweenawan rocks (Pesonen and Halls, 1983). The intrusion of the Logan sills in our sampling area is the likely cause of the BMR and SHR components. The GMN and SMN components may have been acquired during late Keweenawan magmatism when the Duluth Complex was emplaced into the North Shore Volcanic Group at 1099 Ma (Paces and Miller, 1993). The change from reverse to normal polarity evidently occurred between 1097.6 ± 3.7 Ma, the age of a 'reverse' flow within the Osler Group, and 1096.2 ± 1.8 Ma, the age of a 'normal' flow near the base of the Portage Lake Volcanics (Davis and Paces, 1990). Poles GMR, GHR and SHN fall near the 1200 Ma part of the ascending arm of the Logan Loop.

The age of 1400 Ma for the older normal polarity part of the APW path (Fig. 9) is based on 1460–1420 Ma key poles of Buchan et al. (2001), and implies a greater age for the preceding part of the path that includes the GHN pole and Sibley Group poles. The GHN component possibly records tectonism or igneous activity related to the initiation of the Sibley Basin, which may have occurred near 1500 Ma (see Section 2).

Our study has shown that the taconite and other facies of the Gunflint Formation in the Thunder Bay area are not suitable palaeomagnetic targets for the isolation of primary directions, because the entire Animikie succession evidently is too riddled with sills. Although we did not sample Animikean argillite and cherty iron-formation on the Slate Islands, we do not regard these rocks as a promising target for the isolation of primary data because of their proximity to Keweenawan basalt (see Section 3.1) and their disruption by a major bolide impact between ~ 350 and 800 Ma (Dressler and Sharpton, 1997).

7.5. Overprinting of iron formations

The present study confirms that iron formations are very vulnerable to chemical remagnetisation. This was a major conclusion of Schmidt and Clark (1994), who studied 2.5-Ga banded iron formation (BIF) from the Hamersley Basin, Western Australia. Their work showed that although the BIF magnetisation appeared to be pre-folding, it was similar to magnetisations of post-folding iron-ores. Because the BIF was deposited some 300 million years before orogenesis, they concluded that the BIF probably had been remagnetised shortly before folding, uplift and ore-formation. A study by us in progress on BIF from the 1.88-Ga Sokoman Formation, Québec, has examples of pre-folding, syn-folding and post-folding magnetisations, all of similar direction when referred to the appropriate palaeo-horizontal.

One of the few iron formations that may not be largely remagnetised is the Frere Formation in the Earaheedy Basin, Western Australia (Williams et al., in press). In that study, the remanence likewise passed a fold test. However, the Frere Formation where we sampled has not been deeply buried (~ 4 km burial + erosion), shows no obvious sign of regional metamorphism, is not intruded by igneous rocks, and the mineralogy of some beds is very mature. While much iron occurs as Fe^{2+} in siderite-bearing units, causing problems with thermal demagnetisation, the iron in associated red shale and red siltstone is dominantly Fe^{3+}. Indeed, this hematitic facies with stable magnetic remanence could be described as an iron-rich red bed facies (Fe_2O_3T mean = 17.68%, range = 6.73–40.90%).

The vulnerability of many iron formations to remagnetisation seems to arise from their immature mineralogy. Just as some red beds possess an immature mineral assemblage and are prone to remagnetisation (Turner, 1980), so too are iron formations. An abundance of Fe^{2+} in minerals that are unstable in the presence of oxidising fluids, coupled with suitable fluid pathways and porosity, evidently lead to such chemical remagnetisation.

7.6. Asymmetric reversals and the Logan Loop

The origin of apparent asymmetric palaeomagnetic reversals displayed by many rock units associated with the North American Midcontinent Rift System has long been a vexing problem. Early studies noted that successions around Lake Superior were generally of reverse polarity near their base and normal polarity near their top. This led to proposals of superchrons and the use of polarity to correlate Keweenawan units across large distances (DuBois, 1962). Furthermore, the reverse polarity directions

always had steeper inclinations than the normal polarity directions. The asymmetry was thought to result from the presence of secondary components (Palmer, 1970), apparent polar wander, or the failure of the Axial Geocentric Dipole (AGD) hypothesis to account for the Precambrian geomagnetic field (Pesonen and Nevanlinna, 1981).

Using conglomerate and contact tests, Palmer et al. (1981) presented evidence against partial remagnetisation. Pesonen and Halls (1983) used palaeointensities to test between APW and non-AGD models. Their work showed that palaeointensity differences expected for the secondary-component and the non-dipole models were not in evidence after reduction of data to a common palaeolatitude, lending qualified support to the AGD hypothesis and APW model. The qualification alluded to the asymmetry of reverse–normal–reverse–normal directions in volcanic rocks at Mamainse Point on the eastern shore of Lake Superior, which in principle could be explained by fault repetition (Palmer, 1970) although geological evidence for such faulting is lacking (Massey, 1979). Subsequently, Ernst and Buchan (1993) examined the asymmetry displayed by the rocks at Mamainse Point and noted that the two reverse groupings are significantly different, which argues against the repetition being fault-generated. Furthermore, they claimed that in the study of Robertson (1973b) the upper reverse direction is not significantly different from the reverse of both the upper and lower normal flows. Ernst and Buchan (1993) concluded that the Mamainse Point data do not necessarily require reversal asymmetry but are consistent with rapid APW during eruption of the lower reverse (R2) flows followed by a quasi-static interval for the younger N2–R1–N1 flows (notation of Robertson, 1973b). Although Ernst and Buchan (1993) did not document their statistical tests, the R1 direction of Robertson (1973b) is based on only two flows, which invalidates comparison tests because the variance cannot be realistically estimated. On the other hand, Palmer (1970) included five flows in his equivalent R1 direction and this is significantly different from both his N1 and N2 directions using the McFadden and Lowes (1981) comparison test: for R1 versus N1′ (where the prime indicates the reverse of N1) the test statistic is 5.690 compared to $F_{0.05}(2,82) = 3.106$, and for R1 versus N2′ (where the prime indicates the reverse of N2) the test statistic is 3.736 compared to the 0.05 cut-off for $F_{0.05}(2,34) = 3.274$.

However, it would seem best to combine the results of Palmer (1970) and Robertson (1973b) before further comparison tests are performed. The mean directions of the combined results are shown in Table 4. Applying the comparison test of McFadden and Lowes (1981) to the six possible combinations shows that R2, N2, R1 and N1 are all significantly different. The greatest differences involve comparing R2 to the (reverse of the) normal fields (R2 versus N1′, $G = 43.872$ cf. $F_{0.05}(2,86) = 3.102$ and R2 versus N2′, $G = 59.045$ cf. $F_{0.05}(2,130) = 3.066$), and the smallest difference results from comparing N1 and N2 ($G = 4.620$ cf. $F_{0.05}(2,120) = 3.070$). Reversal tests (McFadden and McElhinny, 1990) on the N2–R2, R2–N1 and N1–R1 sets are negative in all cases (Table 4). This is to be expected after the negative results obtained with the tests of McFadden and Lowes (1981). Hence, these results partly agree with the assessment of Ernst and Buchan (1993) that the Mamainse Point data can be explained largely by rapid APW followed by a quasi-static interval. However, R1 remains problematic in that it is significantly different from N1′ and N2′, and although different from R2 it does lie between N1/N2 and R2. To avoid special pleading that the pole path backtracks simultaneously with polarity transitions, we need to examine other explanations for the R1 direction. Ernst and Buchan (1993) questioned whether the R1 groupings of Palmer (1970) and Robertson (1973b) average secular variation because they comprise only five and two sites, respectively. This of course may be so, but an even more insidious related problem may be that the directional scatter is underestimated because the flows are serially correlated. McElhinny et al. (1996) discussed the problem in full, in relation to the Pacific ‘dipole window’, and proposed a method to improve the estimation of directional scatter and secular variation. The Keweenawan data are unsuitable for the method because they are not reported sequentially or are not sufficiently numerous. Without further study to specifically answer the question of serial correlation at Mamainse Point, the significance of the steeper R1, compared to N1 and N2, remains open.

Contrary to findings for many Keweenawan volcanic sections, Symons et al. (1994) claimed that Keweenawan-age (1.1 Ga) alkali syenite–carbonatite

complexes in the Kapuskasing Structural Zone east of Lake Superior possess symmetrical reversals. We have carried out reversal tests on their data (Table 4). Of the five reversals tested, only one is positive and then only as a C class reversal, i.e. better than 20° but not 10° of arc. The positive test is for the Nemegosenda alkaline complex (Costanzo-Alvarez et al., 1993), whereas all other tests are indeterminate because the scatter is too great. When the five separate studies are combined in an overall reversal test the result is again only C class. Nevertheless, through the principle of parsimony this may be taken as reasonable evidence for symmetrical reversals at 1.1 Ga.

The greatest apparent asymmetries observed in the present study were in the taconite, between the magnetite N and R and the hematite N and R components. However, it is the asymmetry for the Sibley conglomerate clasts, residing almost entirely in the declination, that is perhaps most informative because such asymmetry is better explained by secular variation or APW rather than by non-dipole effects. Zonal non-dipole effects are undoubtedly real and asymmetry of inclination is suggestive of long-term (zonal) non-dipole effects. However, it is somewhat more difficult to appeal to non-dipole effects to explain declination differences. This is because the non-dipole effects would have to involve sectoral and tesseral fields, with or without zonal fields, and it is thought that these fields in particular are small and their effects easily contained by normal procedures. When dealing with secondary magnetisations that span reversals, it is usual to assume secular variation has been averaged. Without a viable mechanism to explain non-dipole departures from symmetry in declination, the asymmetries observed in this study and at Mamainse Point are at present best explained in terms of APW. The path can be divided into several intervals of distinctly different APW rates. For the interval 1460–1420 Ma, the rate is ~ 5 cm year^{-1} based on the key poles with those ages. There then appears to be a hiatus or quasi-static interval between 1420 and ~ 1250 Ma. With our re-evaluated age for the Sibley Group of ~ 1500 Ma, there would now appear to be no poles between 1400 and 1300 Ma, which suggests that the amount of APW for this interval is unconstrained and the quasi-static interval maybe an illusion. Between 1250 and 1110 Ma, the rate for the ascending arm of the Logan Loop is an unexceptional 4.2 cm year^{-1}. However, depending on which age is taken for the apex of the Logan Loop, that for the Logan sills of 1108 Ma or that for the Abitibi dykes of 1141 Ma, we calculate a high rate of between 9 and 20 cm year^{-1} for the descending arm of the Logan Loop. Symons et al. (1994) gave an overall APW rate of 12 cm year^{-1}, which is consistent with our estimates. The loop, or hairpin, and sharp change of APW rate could signify a period of major plate reorganisation associated with rifting and voluminous volcanism.

8. Conclusions

The magnetisations found by our study of the 1.88-Ga Gunflint Formation in the Thunder Bay area, Ontario, are of a thermo-chemical origin and are carried both by magnetite and hematite. The components carried by each mineral are of normal and reverse polarity, and most of the groupings fail fold tests and all groupings fail reversal tests. A conglomerate test on Gunflint clasts near the base of the overlying Mesoproterozoic Sibley Group also is negative. Furthermore, the declinations of the Gunflint hematite and magnetite components are different, suggesting that the components were acquired at significantly different times. Although anisotropy of magnetic susceptibility data suggest that the Gunflint Formation has not been dynamically metamorphosed, this finding does not rule out pervasive secondary remagnetisation. We conclude that the Gunflint components identified by our study record remagnetisation events that occurred after the Gunflint Formation was deposited.

Several of the directions found in our study are close to those of 1.1-Ga Keweenawan igneous rocks in the Lake Superior region. The intrusion of the Logan sills in our sampling area is the likely cause of some of the remagnetisation, and several other overprint components may have been acquired during late Keweenawan magmatism. The apparent asymmetrical reversals in our data also are reminiscent of those commonly observed in Keweenawan rocks. After comparing the asymmetry in our data with that in Keweenawan volcanic rocks at Mamainse Point, performing reversal tests on the Mamainse Point data and considering the possibility that serial correlation may partially screen directional scatter, we believe

that the most parsimonious explanation is an interval of rapid apparent polar wander followed by an interval of slower apparent polar wander. Symons et al. (1994) and Buchan et al. (2001) reached similar conclusions. Nevertheless, the reversals at Mamainse Point remain statistically distinguishable. Further work is required on this succession to determine the extent of serial correlation and thus resolve the issues of reversal asymmetry and non-dipole components of the Keweenawan geomagnetic field.

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Appendix A. Sampling localities and details

Site 1. Lower Algal Chert of the Gunflint Formation exposed in the bed of the Whitefish River 2.9 km west of Nolalu, at 48°17.1′N, 270°09.4′E (Franklin and Kustra, 1970, p. 59; Kustra et al., 1977, p. 16). Three block samples contained hematitic ooliths.

Site 3. Banded facies from the Upper Chert Carbonate of the Gunflint Formation exposed in a railcut 1 km south of Blende Lake, at 48°35.2′N,271°13.4′E (Purucker, 1983). Localised folding and intense brecciation of the Gunflint Formation in this area have been ascribed to intrusion of the Logan sills (Moorhouse, 1960; Purucker, 1982, 1983). Three block samples were taken from each limb of a north-trending anticline exposed over a distance of 40 m and a height of 7 m.

Site 4. Calcareous grainstones from the Upper Chert Carbonate of the Gunflint Formation near Boulevard Lake, Thunder Bay, at 48°28.0′N,270°48.2′E (Franklin and Kustra, 1972, p. 34; Kustra et al., 1977, p. 19; Pufahl and Fralick, 2000). Three block samples were taken by the bank of the Current River.

Sites 5–7. Wavy banded, red, jaspilitic and greenalite-bearing grainstone 15–30 cm thick, interbedded with shale, from the Upper Taconite of the Gunflint Formation exposed in a quarry 100 m west of Belrose Road, Thunder Bay, at 48°25.5′N,270°42.2′E (Franklin and Kustra, 1972, p. 34; Kustra et al., 1977, p. 18). The outcrop is capped by a Logan diabase sill. Five block samples and two core samples of jaspilitic grainstone were collected from 1 to 4 m below the base of the sill.

Sites 8–12. Red, jaspilitic and greenalite-bearing grainstone from the Upper Taconite of the Gunflint Formation exposed in the quarry of Towland–Hewitson Construction Limited, 1340 Oliver Road, Thunder Bay, at 48°25.4′N,270°42.7′E. Fifteen block samples were collected. No diabase sill is evident here.

Sites 2, 13–22. Wavy banded, red, hematitic and dolomitic grainstone, locally folded and brecciated, from the Lower East Taconite of the Gunflint Formation exposed in a road cut on the Trans-Canadian Highway 1 km south of Loon Lake, at 48°37.6′N,271°14.7′E (Franklin and Kustra, 1970, p. 63; Franklin and Kustra, 1972, p. 36). Thirty-eight samples (15 core, 23 block) were collected from the limbs of several small folds.

Site 23. Polymict basal conglomerate (Loon Member) of the Pass Lake Formation at the base of the Sibley Group exposed in a railcut at Pass Lake, 48°34.0′N,271°16.2′E (Kustra et al., 1977, p. 25; Franklin et al., 1980; Cheadle, 1986). Block samples were collected from 20 pebbles and cobbles of peloidal and banded taconite derived from the Gunflint Formation.

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